Abstract

[To write: Summarize the problem, implementation, datasets, validation approach, and main measured findings in about 150–250 words. Write this last.]

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1 Introduction

[To write: Introduce frequent itemset mining, explain the project goals, and summarize the implementation and evaluation contributions.]

2 Background and Algorithm

2.1 Problem Definition

[To write: Define transactions, items, itemsets, absolute and relative support, and the minimum support threshold. State how a fractional threshold is converted to a transaction count.]

2.2 FP-Tree Construction

[To write: Explain frequency counting, removal of infrequent items, deterministic item ordering, prefix sharing, and the header table. Cite the original algorithm using `\citep{han2000fpgrowth}`*.]*

2.3 Recursive Pattern Mining

[To write: Explain conditional pattern bases, conditional FP-trees, recursion, stopping conditions, and any single-path optimization implemented. Include pseudocode or a small tree diagram where helpful.]

2.4 Computational Considerations

[To write: Discuss tree size, recursion, output size, and factors that affect runtime and memory. Distinguish theoretical expectations from measurements.]

3 Datasets and Preprocessing

3.1 Benchmark Data

[To write: Describe the UCI Gas Sensor Array Drift Dataset and any other benchmarks actually used. Cite the dataset with `\citep{vergara2012gas}`*. Record transaction counts, item counts, transaction lengths, and the selected batches or subsets.]*

3.2 Transaction Encoding

Read datasheet
- Clean Data
- Encode items
- Count item frequencies

Remove infrequent items
Sort remaining items in each transaction by frequency
Build FP-Tree

[To write: Document selected features, normalization, binning rules, threshold fitting, item-ID mapping, and treatment of gas labels and batches. Explain how continuous measurements become discrete transactions.]

4 Implementation

4.1 Program Structure and Data Structures

[To write: Describe the language, main modules, node representation, header links, support counts, memory ownership, and output format.]

4.2 Input Handling and Design Choices

[To write: Document input assumptions, duplicate-item handling, tie breaking, parameter validation, and any implementation optimizations.]

4.3 CUDA Implementation

5 Correctness Validation

5.1 Worked Example

[To write: Show a small transaction database and minimum support. List the expected frequent itemsets and exact support counts, illustrate the FP-tree and at least one conditional tree, and compare actual output.]

5.2 Additional Tests

[To write: Report the tests actually run and their outcomes. Consider empty input, repeated items, identical transactions, support boundaries, no frequent items, tied frequencies, and comparison with exhaustive enumeration on small databases.]

6 Experimental Methodology

6.1 Hardware and Software

[To write: Record CPU, RAM, operating system, compiler version and flags, and GPU/CUDA details if applicable. Include commands needed to reproduce runs.]

6.2 Measurements and Experimental Parameters

[To write: Specify support thresholds, input sizes, repetitions, warm-up policy, timer, and memory measurement method. Define whether timings include preprocessing, input/output, tree construction, mining, and GPU transfers. State how variability and incomplete runs are reported.]

7 Results and Discussion

7.1 Execution Time

[To write: Present measured runtime against minimum support and input size, with units and variability. Explain observed trends.]

7.2 Memory Usage and Output Size

[To write: Report peak memory and number of frequent itemsets for the same configurations. Explain the relationship between support, tree size, output size, and resource use.]

7.3 Limitations

[To write: Discuss dataset and preprocessing choices, measurement limits, scalability, and any failed or incomplete experiments.]

8 Conclusion

[To write: Summarize what was implemented, the evidence for correctness, the main measured results, and specific future improvements.]

References

[1] Jiawei Han, Jian Pei, and Yiwen Yin. Mining frequent patterns without candidate generation. In *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*, pages 1–12. Association for Computing Machinery, 2000. doi: 10.1145/342009.335372. URL https://doi.org/10.1145/342009.335372.

[2] Alexander Vergara. Gas Sensor Array Drift Dataset. UCI Machine Learning Repository, 2012. URL https://doi.org/10.24432/C5RP6W. Dataset.

A Reproducibility Details

[To write: Include exact build and run commands, preprocessing parameters, and additional validation or result tables that would interrupt the main text. Remove this appendix if all details fit in the main report.]